

WRENTHAM AGDM, AB, Canada

WMO: 713430

Lat: 49.4952N

Lon: 112.1116W

Elev: 3097

StdP: 13.12

Time Zone: -7.00 (NAM)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-16.6	-10.6	-22.6	1.8	-16.3	-16.3	2.5	-10.3	36.5	35.6	33.8	36.8	7.2	290	0.725

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	27.3	88.4	60.7	85.0	60.3	81.6	59.5	65.4	78.9	63.4	78.1	61.5	76.9	9.9	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.6	88.7	71.8	58.0	80.7	69.1	55.8	74.4	66.4	31.9	78.5	30.4	77.9	28.9	77.1	75.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 31.6	27.8	24.5	DB	-24.5	94.6	6.5	3.5	-29.2	97.1	-33.0	99.2	-36.6	101.1	-41.3	103.6	(4)
(5)			WB	-24.8	70.1	6.4	3.1	-29.5	72.3	-33.2	74.1	-36.8	75.8	-41.5	78.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	42.6	23.1	21.5	32.6	42.3	51.6	58.8	65.0	64.1	55.2	43.3	30.9	21.2	(6)
(7)		DBStd	19.10	14.44	14.80	12.68	8.91	8.27	5.88	5.15	6.38	8.75	10.19	13.28	14.66	(7)
(8)		HDD50	4276	835	797	543	254	81	4	0	1	44	248	577	892	(8)
(9)		HDD65	8344	1300	1217	1005	680	419	202	65	95	310	672	1022	1357	(9)
(10)		CDD50	1573	0	0	3	24	131	267	464	437	200	42	5	0	(10)
(11)		CDD65	165	0	0	0	0	4	15	64	66	16	0	0	0	(11)
(12)		CDH74	3202	0	0	1	14	151	279	1132	1220	388	17	0	0	(12)
(13)		CDH80	1042	0	0	0	1	27	51	378	459	125	1	0	0	(13)
(14)	Wind	WSAvg	11.5	13.5	11.4	12.5	12.3	11.7	11.0	8.7	8.8	10.5	12.0	12.7	12.6	(14)
(15)	Precipitation	PrecAvg	14.0	0.4	0.4	0.7	1.3	1.9	3.3	1.2	1.3	1.4	0.7	0.5	0.5	(15)
(16)		PrecMax	20.6	0.7	0.8	1.5	2.4	4.0	9.9	2.6	3.0	3.3	2.2	1.3	0.9	(16)
(17)		PrecMin	8.0	0.2	0.2	0.2	0.5	0.6	0.4	0.0	0.2	0.2	0.2	0.2	0.2	(17)
(18)		PrecStd	3.2	0.1	0.2	0.3	0.6	1.0	2.2	0.8	0.8	0.9	0.5	0.3	0.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.9	55.8	64.4	75.7	83.4	85.5	92.3	93.0	88.9	76.4	64.8	52.2	(19)
MCWB			42.7	42.1	46.3	51.8	56.6	63.3	62.3	60.7	58.5	53.7	47.8	40.4	(20)	
2%		DB	47.3	48.4	59.1	69.4	78.6	80.6	87.8	89.0	83.8	69.5	56.5	46.4	(21)	
		MCWB	39.4	38.5	44.4	49.1	54.9	60.0	62.5	60.1	56.7	51.1	44.3	38.1	(22)	
5%		DB	43.6	43.9	54.5	63.7	73.3	77.1	84.3	85.3	78.4	63.6	51.8	42.9	(23)	
		MCWB	36.7	36.0	42.1	46.4	52.8	58.8	62.5	59.2	55.3	48.4	41.8	36.1	(24)	
10%		DB	40.5	40.3	49.9	58.4	68.6	73.2	80.6	81.4	72.7	58.4	47.4	39.5	(25)	
		MCWB	35.2	34.2	39.7	43.6	50.7	57.1	62.1	58.6	53.4	45.9	39.1	33.9	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.6	42.9	47.8	52.9	58.3	66.9	70.6	65.7	60.8	55.3	49.4	42.4	(27)
MCD B			51.8	52.8	61.4	71.9	79.5	79.0	82.4	79.6	78.1	73.2	61.0	49.7	(28)	
2%		WB	39.7	39.0	45.3	49.8	55.6	63.1	66.5	63.0	58.2	52.0	45.2	38.7	(29)	
		MCD B	46.7	47.4	57.5	67.1	74.0	76.6	79.5	80.7	78.7	66.6	55.6	45.6	(30)	
5%		WB	37.3	36.5	42.6	47.3	53.8	60.2	64.6	61.3	56.3	49.2	42.3	36.3	(31)	
		MCD B	42.9	43.4	53.4	61.8	70.6	73.4	78.4	79.6	76.0	62.1	51.2	42.3	(32)	
10%		WB	35.2	34.3	39.9	44.5	51.9	58.1	62.9	59.9	54.2	46.7	39.5	34.1	(33)	
		MCD B	40.2	40.0	49.1	56.6	65.8	70.7	77.2	77.5	71.4	57.1	46.9	39.6	(34)	
(35)	Mean Daily Temperature Range	MDBR	18.1	19.5	20.8	22.9	22.8	22.3	27.3	28.0	25.0	22.1	19.4	17.7	(35)	
5% DB		MCD BR	18.6	22.4	27.3	32.3	31.1	29.0	34.1	36.1	34.7	28.8	24.4	19.5	(36)	
		MCWBR	12.8	14.5	15.4	15.6	12.8	13.4	14.8	13.2	13.8	14.1	14.5	13.3	(37)	
		MCD BR	18.3	22.0	26.0	30.1	28.0	26.0	29.1	30.6	32.4	27.8	23.8	19.3	(38)	
(39)		5% WB	MCWBR	13.1	15.1	15.5	15.2	12.3	13.5	14.7	12.6	13.3	14.0	14.5	13.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.256	0.264	0.285	0.323	0.338	0.336	0.338	0.355	0.317	0.285	0.258	0.244	(40)	
taud		2.462	2.488	2.486	2.382	2.387	2.441	2.445	2.367	2.450	2.522	2.503	2.446	(41)		
Ebn at Noon		262	288	298	293	292	292	290	279	281	274	259	251	(42)		
Edh at Noon		19	23	28	35	36	35	34	35	29	22	18	16	(43)		
(44)	All-Sky Solar Radiation	RadAvg	400	718	1120	1548	1839	2049	2247	1792	1272	796	448	317	(44)	
(45)		RadStd	34	55	89	105	150	140	117	111	103	78	32	29	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)			
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (60 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page